

Academic Positions

Columbia University

Founder's Postdoctoral Fellow, Department of Statistics

New York, NY

2025–present

Education

University of Pennsylvania

Ph.D. in Statistics, The Wharton School

Advisor: T. Tony Cai

Philadelphia, PA

2020–2025

Indian Statistical Institute

M.Stat., specialization in Probability

Ranked first; Best master's dissertation; First Division with Distinction

Kolkata, India

2018–2020

Indian Statistical Institute

B.Stat. (Honours)

Ranked first; First Division with Distinction

Kolkata, India

2015–2018

Publications

Author notation: ($\alpha\beta$) Alphabetical ordering. * Equal contribution.

1. ($\alpha\beta$) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Optimal federated learning for functional mean estimation under heterogeneous privacy constraints. In press, *Journal of the Royal Statistical Society: Series B*, 2026. [paper]
2. ($\alpha\beta$) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Optimal federated learning for nonparametric regression with heterogeneous distributed differential privacy constraints. In press, *Journal of the American Statistical Association*, 2026. [paper]
3. **Abhinav Chakraborty**, Sayak Chatterjee, and Sagnik Nandy. PriME: Privacy-aware membership profile estimation in networks. In press, *SIAM Journal on Mathematics of Data Science*, 2026. [paper]
4. ($\alpha\beta$) T. Tony Cai, **Abhinav Chakraborty**, and Yichen Wang. Optimal differentially private ranking from pairwise comparisons. *Journal of the American Statistical Association*, 2026. [paper]
5. ($\alpha\beta$) Arnab Auddy, T. Tony Cai, and **Abhinav Chakraborty**. Minimax and adaptive transfer learning for nonparametric classification under distributed differential privacy constraints. *Journal of the Royal Statistical Society: Series B*, 88(3):903–929, 2026. [paper]
6. **Abhinav Chakraborty**, Arnab Auddy, and T. Tony Cai. When data can't meet: Estimating correlation across privacy barriers. *Advances in Neural Information Processing Systems*, 38, 2025. (Spotlight) [paper]
7. Ziang Niu*, **Abhinav Chakraborty***, Oliver Dukes, and Eugene Katsevich. Reconciling model-X and doubly robust approaches to conditional independence testing. *The Annals of Statistics*, 52(3):895–921, 2024. [paper]
8. **Abhinav Chakraborty***, Anirban Chatterjee*, and Abhinandan Dalal*. PrIsing: Privacy-preserving peer effect estimation via Ising model. *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, PMLR 238:2692–2700, 2024. [paper]

Papers Under Revision

1. ($\alpha\beta$) T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. The cost of adaptation under differential privacy: Optimal adaptive federated density estimation. arXiv preprint arXiv:2512.14337, 2025. Under major revision at *The Annals of Statistics*. [paper]
2. Sayak Chatterjee, Anirban Chatterjee, **Abhinav Chakraborty**, and Bhaswar B. Bhattacharya. Asymptotic normality of subgraph counts in sparse inhomogeneous random graphs. arXiv preprint arXiv:2512.12937, 2025. Under minor revision at *Bernoulli*. [paper]

3. **Abhinav Chakraborty***, Yuetian Luo*, and Rina Foygel Barber. Stability and accuracy trade-offs in statistical estimation. arXiv preprint arXiv:2601.11701, 2026. Under major revision at *IEEE Transactions on Information Theory*. [paper]
4. $(\alpha\beta)$ T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Federated nonparametric hypothesis testing with differential privacy constraints: Optimal rates and adaptive tests. arXiv preprint arXiv:2406.06749, 2024. R&R at *JMLR*. [paper]

Preprints

1. Lucas Kania*, **Abhinav Chakraborty***, Edward Kennedy, Larry Wasserman, and Sivaraman Balakrishnan. Optimal inference with black-box predictions. arXiv preprint arXiv:2608.10155, 2026. [paper]
2. **Abhinav Chakraborty** and Sagnik Nandy. Transfer learning in high-dimensional clustering: Minimax thresholds and applications in single-cell data. arXiv preprint arXiv:2607.25031, 2026. [paper]
3. **Abhinav Chakraborty** and Subha Maity. The statistical cost of adaptation in multi-source transfer learning. arXiv preprint arXiv:2605.09471, 2026. [paper]
4. **Abhinav Chakraborty***, Junu Lee*, and Eugene Katsevich. Comparing three learn-then-test paradigms in a multivariate normal means problem. arXiv preprint arXiv:2601.07764, 2026. [paper]
5. **Abhinav Chakraborty***, Jeffrey Zhang*, and Eugene Katsevich. Doubly robust and computationally efficient high-dimensional variable selection. arXiv preprint arXiv:2409.09512, 2024. [paper]
6. **Abhinav Chakraborty**, Soumendu Sundar Mukherjee, and Arijit Chakrabarti. High dimensional PCA: A new model selection criterion. arXiv preprint arXiv:2011.04470, 2020. [paper]

Awards and Honors

IMS Travel Award Frontiers in Statistical Machine Learning Workshop.	2025
Travel Award, Optimization and Statistical Learning Workshop Columbia University.	2025
Lawrence D. Brown Best Student Paper Award The Wharton School, University of Pennsylvania.	2024
George James Term Award Institute of Mathematical Statistics Annual Meeting.	2022, 2024
ISI Alumni Association J. K. Ghosh Memorial Gold Medal Top scorer in the M.Stat. program.	2020
P. C. Mahalanobis Gold Medal Best master's dissertation; outstanding M.Stat. student.	2020
ISI Alumni Association Mrs. M. R. Iyer Memorial Gold Medal Top scorer in the B.Stat. program.	2018
Nikhilesh Bhattacharya Gold Medal Highest marks in Statistics in the B.Stat. (Hons.) program.	2018
INSPIRE Scholarship Awarded to students in the top 1% of the Indian School Certificate Examination.	2015
Indian National Mathematics Olympiad qualifier Top-30 candidate in the Regional Mathematics Olympiad, 2013.	2014

Talks

MCP Conference, Philadelphia, PA Invited talk: Learn-then-test procedures.	2025
IISA Conference, Lincoln, NE Invited talk: Federated transfer learning.	2025
IMS Annual Meeting, Bochum, Germany Federated nonparametric regression.	2024
Ph.D. Student Seminar, University of Pennsylvania Optimal distributed private estimators.	2023

IMS Annual Meeting, London, UK

2022

High dimensional PCA.

P. C. Mahalanobis Gold Medal Presentation, Indian Statistical Institute

2020

Model selection in high dimensional PCA.

D. Basu Memorial Presentation, Indian Statistical Institute

2018

MCMC methods in Bayesian inference.

Teaching and Mentoring

Teaching Assistant, The Wharton School, University of Pennsylvania

2021–2024

Introduction to Probability (STAT 4300); Probability Theory (MATH 6480); Mathematical Statistics (STAT 971); Applied Econometrics (STAT 520).

Ph.D. Peers Program, The Wharton School, University of Pennsylvania

2023

Mentored an incoming Wharton Ph.D. student in navigating the program and transitioning to Penn.

Professional Service

Journal reviewer: *The Annals of Statistics*; *Journal of the American Statistical Association (JASA)*; *IEEE Transactions on Information Theory*; *Journal of the Royal Statistical Society: Series B (JRSSB)*; and *Biometrika*.